

T-verse Module: Automated Vendor Anomaly & Quality Tracker

1. Feature Brief

The Vendor Anomaly & Quality Tracker is a passive intelligence module within the T-verse ERP. It automatically evaluates the reliability of suppliers (printers, fabric manufacturers, packaging vendors) by tracking two primary metrics: **Delivery Reliability** (On-Time In-Full or OTIF) and **Defect Rate** (Quality Control rejections). Instead of a procurement manager manually reviewing spreadsheets, this system maintains a dynamic "Vendor Trust Score." If a vendor's score drops below a safe threshold due to late shipments or high defect rates, the system alerts the team and recommends backup suppliers.

2. System Architecture & Workflow

- **Data Intake (The Trigger):** Every time a warehouse worker logs a received Purchase Order (PO), the system calculates the delta between the Promised_Date and Actual_Receipt_Date.
- **Quality Control (QC) Loop:** When the QA team flags damaged items (e.g., misaligned prints, fabric tears), they log the defect against the specific batch ID. The backend maps this defect directly back to the originating vendor.
- **The Scoring Engine:** A scheduled Spring Boot cron job runs weekly, crunching the historical data to recalculate each vendor's Trust Score (0-100 scale).
- **Alerts & Mitigation:** If a score drops below 75, an automated email is triggered to the supply chain team outlining the specific anomalies (e.g., "Vendor X has been 3 days late on the last 4 shipments").

3. Tech Stack Integration (T-verse)

Integrating this into the com.thalasi.tverse package involves heavy data aggregation.

- **Backend (Java 25 & Spring Boot 3.4):** Utilize Spring's @Scheduled annotation for the weekly batch processing. Java 25 Records are perfect here for creating immutable, lightweight Data Transfer Objects (DTOs) when aggregating millions of rows of PO and QC data.
- **Database Optimization:** This requires complex SQL GROUP BY and window functions

to calculate trailing 30-day and 90-day averages efficiently without locking up the main transactional database.

- **Frontend (React):** A dedicated "Vendor Command Center" dashboard utilizing charting libraries (like Recharts or Chart.js) to visualize a supplier's performance degradation over time.

4. AI Integrations

4.1 Natural Language QC Categorization

Often, QA workers just type unstructured notes when rejecting items (e.g., "the white border is messed up" or "print is bleeding"). An NLP agent parses these raw text logs and categorizes them into standardized defect codes (e.g., DEFECT_PRINT_ALIGNMENT, DEFECT_FABRIC_TEAR). This allows the system to identify if a vendor has a specific, recurring issue.

4.2 Risk Forecasting

An AI model looks at macro-trends. If a specific fabric supplier is historically late during the monsoon season, the AI proactively suggests padding the lead time by 7 days for any purchase orders placed in June or July.

5. Real-Life Business Impact

- **Protects Launch Timelines:** Identifying a degrading supplier *before* placing a massive order for a new collection ensures the brand doesn't miss crucial marketing launch dates.
- **Negotiation Power:** Provides concrete, data-backed dashboards that procurement can use to negotiate discounts or penalties with underperforming vendors.
- **Automated Accountability:** Removes human bias from vendor relationships, relying strictly on empirical performance data.